

Lab - Azure CLI - Azure Web Apps

To Begin with the Lab

Summary

This lab demonstrates how to use the Azure CLI to create, deploy, manage, and delete an Azure Web App. You will create a Resource Group, App Service Plan, Web App, deploy a sample application, verify the deployment, and clean up Azure resources using Azure CLI commands.

Understanding

Azure CLI is a command-line tool that allows you to manage Azure resources from Windows, macOS, and Linux. Instead of using the Azure Portal, you can automate tasks such as creating Resource Groups, deploying Web Apps, managing App Service Plans, and deleting resources through simple commands.

Azure Web Apps is a Platform as a Service (PaaS) offering that enables developers to host web applications without managing servers or operating systems. Azure automatically handles scaling, patching, monitoring, and infrastructure maintenance.

Example:

A developer wants to deploy an ASP.NET Core application using only the command line. They use Azure CLI to log in, create a Resource Group, create an App Service Plan, create a Web App, deploy the application, verify the website, and finally remove all resources after testing.

Lab Steps

Step 1: Install Azure CLI

- Download and install Azure CLI on your computer.
- Open Command Prompt, PowerShell, or Terminal.
- Verify the installation using:
- Run this

<code>az --version</code>

```
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PS /home/ritesh> az --version
azure-cli 2.87.0 *

core 2.87.0 *
telemetry 1.1.0

Extensions:
ml 2.43.0
ssh 2.0.9

Dependencies:
msal 1.36.0
azure-mgmt-resource 24.0.0

Python location '/usr/bin/python3.12'
Config directory '/home/ritesh/.azure'
Extensions directory '/home/ritesh/.azure/cliextensions'
Extensions system directory '/usr/lib/python3.12/site-packages/azure-cli-extensions'

Python (Linux) 3.12.9 (main, Jun 2 2026, 15:06:48) [GCC 13.2.0]

Legal docs and information: aka.ms/AzureCLILegal

You have 2 update(s) available. They will be updated with the next build of Cloud Shell.
PS /home/ritesh>
```

Step 2: Login to Azure

- Sign in to your Azure account.
- Run:

```
az login
```

- A browser window opens for authentication.
- Sign in successfully.

Step 3: Select Azure Subscription (Optional)

- View available subscriptions:

```
az account list --output table
```

- Set the required subscription:

```
az account set --subscription "<Subscription Name>"
```

```
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PS /home/ritesh> az account list --output table
Name CloudName SubscriptionId TenantId State IsDefault
-----
Visual Studio Enterprise Subscription AzureCloud 294539c9-c109-4db8-b785-0043716c0b11 3d18f3ae-8875-4771-aaa9-a9afcd43d751 Enabled True
PS /home/ritesh>
```

Step 4: Create a Resource Group

- Create a new Resource Group.

```
$location = "East US"
```

```
$rg = "Webapp"
```

```
$app = "Webapp-eus-13"
```

```
az group create --name $rg --location $location
```

```
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PS /home/ritesh> $location = "East US"
PS /home/ritesh> $rg = "Webapp"
PS /home/ritesh> $app = "Webapp-eus-13"
PS /home/ritesh> az group create --name $rg --location $location

^[[{
  "id": "/subscriptions/294539c9-c109-4db8-b785-0043716c0b11/resourceGroups/Webapp",
  "location": "eastus",
  "managedBy": null,
  "name": "Webapp",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}]
```

5: Create an App Service Plan

- Create a Linux App Service Plan.
- Run this Command.

az appservice plan create --name **WebappAppServicePlan** --resource-group **Webapp** --sku **B1** --is-linux

```
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PS /home/ritesh> az appservice plan create --name WebappAppServicePlan --resource-group Webapp --sku B1 --is-linux
Creating App Service Plan 'WebappAppServicePlan' (Linux).
{
  "asyncScalingEnabled": false,
  "elasticScaleEnabled": false,
  "freeOfferExpirationTime": "2026-08-16T08:28:48.6433333",
  "geoRegion": "East US",
  "hyperV": false,
  "id": "/subscriptions/294539c9-c109-4db8-b785-0043716c0b11/resourceGroups/Webapp/providers/Microsoft.Web/serverfarms/WebappAppServicePlan",
  "isCustomMode": false,
  "isSpot": false,
  "isXenon": false,
  "kind": "linux",
  "location": "eastus",
  "maximumElasticWorkerCount": 1,
  "maximumNumberOfWorkers": 0,
  "name": "WebappAppServicePlan",
  "numberOfSites": 0,
  "numberOfWorkers": 1,
  "perSiteScaling": false,
  "provisioningState": "Succeeded",
  "reserved": true,
  "resourceGroup": "Webapp",
  "sku": {
    "capacity": 1,
    "family": "B",
    "name": "B1",
    "size": "B1",
    "tier": "Basic"
  },
  "status": "Ready",
  "subscription": "294539c9-c109-4db8-b785-0043716c0b11",
  "targetWorkerCount": 0,
  "targetWorkerSizeId": 0,
  "type": "Microsoft.Web/serverfarms",
  "zoneRedundant": false
}
```

Step 6: Create an Azure Web App

- Create a Web App using a runtime.

az webapp create --resource-group **Webapp** --plan **WebappAppServicePlan** --name **Webapp-dev-eus-12** --runtime "DOTNETCORE:8.0"

```
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PS /home/ritesh> az webapp create --resource-group Webapp --plan WebappAppServicePlan --name Webapp-dev-eus-12 --runtime "DOTNETCORE:8.0"
Webapp 'Webapp-dev-eus-12' created. Deploy your code with: az webapp deploy
{
  "autoGeneratedDomainNameLabelScope": null,
  "availabilityState": "Normal",
  "clientAffinityEnabled": true,
  "clientAffinityPartitioningEnabled": null,
  "clientAffinityProxyEnabled": false,
  "clientCertEnabled": false,
  "clientCertExclusionPaths": null,
  "clientCertMode": "Required",
  "cloningInfo": null,
  "containerSize": 0,
  "customDomainVerificationId": "018BEC2E058DC592DE1A69B232E70A28543C2565A7CF6738EB11C1CD6FEBD97B",
  "dailyMemoryTimeQuota": 0,
  "daprConfig": null,
  "defaultHostName": "webapp-dev-eus-12.azurewebsites.net",
  "dnsConfiguration": {
    "dnsAltServer": null,
    "dnsLegacySortOrder": null,
    "dnsMaxCacheTimeout": null,
    "dnsRetryAttemptCount": null,
    "dnsRetryAttemptTimeout": null,
    "dnsServers": null
  },
  "enabled": true,
  "enabledHostNames": [
    "webapp-dev-eus-12.azurewebsites.net",
    "webapp-dev-eus-12.scm.azurewebsites.net"
  ],
  "endToEndEncryptionEnabled": false,
  "extendedLocation": null,
  "functionAppConfig": null,
  "hostNameSslStates": [
    {
      "hostType": "Standard",
      "name": "webapp-dev-eus-12.azurewebsites.net",
      "sslState": "Disabled",
    }
  ]
}
```

- Update pricing tier (service plan).
- Run this Command.

```
az appservice plan update --name WebappAppServicePlan --resource-group Webapp --sku S1
```

```
PS /home/ritesh> az appservice plan update --name WebappAppServicePlan --resource-group Webapp --sku S1
{
  "asyncScalingEnabled": false,
  "elasticScaleEnabled": false,
  "extendedLocation": null,
  "freeOfferExpirationTime": "2026-08-16T08:28:00",
  "geoRegion": "East US",
  "hostingEnvironmentProfile": null,
  "hyperV": false,
  "id": "/subscriptions/294539c9-c189-4db8-b785-0043716c0b11/resourceGroups/Webapp/providers/Microsoft.Web/serverFarms/WebappAppServicePlan",
  "identity": null,
  "installScripts": null,
  "isCustomMode": false,
  "isSpot": false,
  "isXenon": false,
  "kind": "linux",
  "kubeEnvironmentProfile": null,
  "location": "East US",
  "maximumElasticWorkerCount": 1,
  "maximumNumberOfWorkers": 10,
}
```

Step 6: Create a Deployment Slot

- Create a **staging** deployment slot.

```
az webapp deployment slot create --name Webapp-dev-eus-12 --resource-group Webapp --slot staging
```

```
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PS /home/ritesh> az webapp deployment slot create --name Webapp-dev-eus-12 --resource-group Webapp --slot staging
{
  "autoGeneratedDomainNameLabelScope": null,
  "availabilityState": "Normal",
  "clientAffinityEnabled": true,
  "clientAffinityPartitioningEnabled": null,
  "clientAffinityProxyEnabled": false,
  "clientCertEnabled": false,
  "clientCertExclusionPaths": null,
  "clientCertMode": "Required",
  "cloningInfo": null,
  "containerSize": 0,
  "customDomainVerificationId": "0188EC2E058DC592DE1A69B232E70A28543C2565A7CF6738EB11C1CD6FEBD97B",
  "dailyMemoryTimeQuota": 0,
  "daprConfig": null,
  "defaultHostName": "webapp-dev-eus-12-staging.azurewebsites.net",
  "dnsConfiguration": {
    "dnsAltServer": null,
    "dnsLegacySortOrder": null,
    "dnsMaxCacheTimeout": null,
    "dnsRetryAttemptCount": null,
    "dnsRetryAttemptTimeout": null,
    "dnsServers": null
  },
  "enabled": true,
  "enabledHostNames": [
    "webapp-dev-eus-12-staging.azurewebsites.net",
    "webapp-dev-eus-12-staging.scm.azurewebsites.net"
  ],
  "endToEndEncryptionEnabled": false,
  "extendedLocation": null,
  "functionAppConfig": null,
  "hostNameSslStates": [
    {
      "hostType": "Standard",
      "name": "webapp-dev-eus-12-staging.azurewebsites.net",

```

Step 7: Verify Deployment Slots

- Go to just Created Webapp
- Navigate to **Deployment slot**
- List all deployment slots.

Home > Resource Manager | Resource groups > Webapp > Webapp-dev-eus-12

Webapp-dev-eus-12 Deployment slots ☆ ...

Web App

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Deployment

Deployment slots

Deployment Center

Deployment slots are live apps with their own hostnames. App content and configurations elements can be swapped between two deployment slots, including the production slot.

Name	Status	App service plan	Traffic %
webapp-dev-eus-12 PRODUCTION	Running	WebappAppServicePlan	100
Webapp-dev-eus-12-staging	Running	WebappAppServicePlan	0